Electron microscopy and microanalysis

The light microscope cannot see things smaller than about a quarter of a micrometre, because the wavelength of light sets the limit. Electrons have a wavelength tens of thousands of times shorter. The electron microscope can therefore show crystal lattices and individual columns of atoms – and when the electrons strike the specimen, it emits X-rays that reveal which elements it consists of. The page has an interactive resolution limit and a simulation of electron trajectories in the specimen.

> DTH, course 1848 (1987), 6 points: Electron microscopy and microanalysis. The course combined theory with work at the microscopes – from specimen preparation to images and spectra.

Electrons as waves

Louis de Broglie proposed in 1924 that particles have a wavelength $\lambda = h/p$. Clinton Davisson and Lester Germer in the USA and George Paget Thomson in Aberdeen showed in 1927 that electrons are diffracted by crystals – Thomson, whose father had shown that the electron is a particle. An electron accelerated through the voltage U has the wavelength

$$\lambda = \frac{h}{\sqrt{2 m_e e U \left(1 + \frac{eU}{2 m_e c^2}\right)}}$$

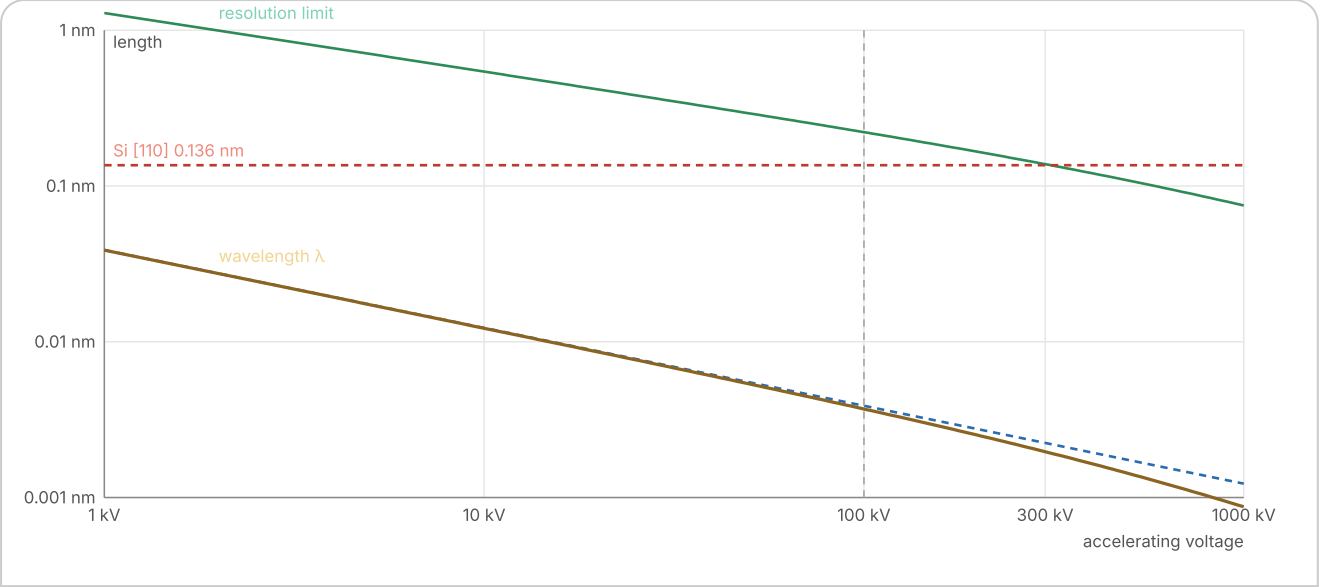
where the bracket is the relativistic correction, already 10 % at 100 kV. At 100 kV $\lambda = 3.7$ pm, more than 100 000 times shorter than green light. The resolution is therefore limited not by the wavelength but by the aberrations of the lenses.

Magnetic lenses

Hans Busch showed in 1926 that a rotationally symmetric magnetic field from a coil focuses electrons as a glass lens focuses light. The focal length can be changed with the current. But Otto Scherzer proved in 1936 that such lenses always have positive spherical aberration: electrons far from the axis are bent too strongly. One cannot, as with glass, combine a converging lens with a diverging one. So small aperture angles α must be used, and the best resolution becomes a compromise between diffraction ($\propto \lambda/\alpha$) and spherical aberration ($\propto C_s \alpha^3$):

$$d_{\min} \approx 0.66\, C_s^{1/4} \lambda^{3/4}$$

With C_s of a couple of millimetres this gives about 0.3 nm at 100 kV – good enough to see crystal lattices, but only just.



The transmission microscope

Ernst Ruska and Max Knoll built the first electron microscope in Berlin in 1931; in 1933 it surpassed the resolution of the light microscope, and Siemens delivered the first commercial one in 1939. Ruska received the Nobel Prize in 1986 – 55 years later. In a transmission electron microscope (TEM) the electrons pass through a specimen thinner than 100 nm, and a series of lenses forms a magnified image on a fluorescent screen or a camera. Contrast comes from scattering: thick and heavy regions scatter more. In crystalline specimens diffraction gives contrast from dislocations and grain boundaries – and if the microscope is set to the back focal plane, one sees the specimen's electron diffraction pattern, which reveals the crystal structure. In high-resolution TEM (HRTEM) the diffracted beams interfere and form an image of the lattice itself.

Specimen preparation is a large part of the work: metal specimens are electropolished or ion-milled thin, biological specimens are embedded, cut into ultra-thin sections and stained with heavy metals.

The scanning microscope

In a scanning electron microscope (SEM) the beam is focused to a small spot that is scanned line by line across the surface of a bulk specimen, like the electron beam in a television picture. For each point a signal is measured, and the image is built up point by point. Manfred von Ardenne built one in 1937, but it was Charles Oatley's group in Cambridge that developed it into the practical

instrument that the Cambridge Instrument Company began selling in 1965. The most important signals are:

- **Secondary electrons** with energies below 50 eV escape only from the top few nanometres. Edges and slopes give more signal, so the image looks like an illuminated surface with enormous depth of field – the famous images of insect eyes and pollen grains.
- **Backscattered electrons** have almost the primary energy and come from deeper. Their number grows with atomic number, so heavy phases look bright – compositional contrast.
- **Characteristic X-rays** from the specimen's atoms, giving the chemical composition (see below).

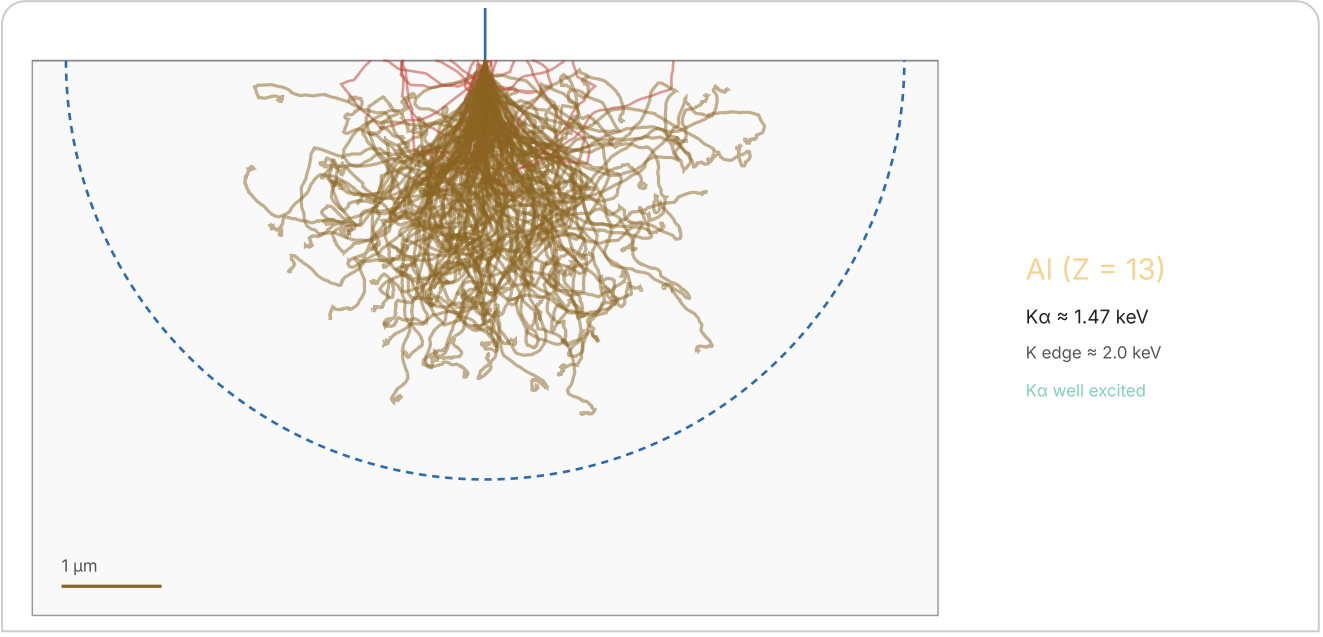
Non-conducting specimens charge up and must be coated with a thin layer of gold or carbon.

Interaction volume

The electrons do not go straight into the specimen. They are scattered elastically by the nuclei and lose energy in inelastic collisions with the electrons, following zigzag paths in a pear-shaped *interaction volume*. Its size can be estimated with Kanaya and Okayama's formula (1972):

$$R = \frac{0.0276\, A\, E_0^{1.67}}{Z^{0.89}\, \rho}\ \mu\text{m} \qquad (A \text{ in g/mol},\ E_0 \text{ in keV},\ \rho \text{ in g/cm}^3)$$

At 20 kV it is about 4 µm in aluminium but only 1 µm in gold. It determines how finely microanalysis can be done: the X-rays come from the whole volume, not from the small spot on the surface. The interaction volume is best understood by simulating many electron trajectories with the Monte Carlo method.



X-ray microanalysis

When an electron in the beam knocks an electron out of the K shell of an atom, an electron from the L shell drops down, and an X-ray photon is emitted with an energy characteristic of the element. Henry Moseley measured the lines of almost all elements in 1913 and found that the square root of the frequency increases linearly with atomic number:

$$\sqrt{f_{K\alpha}} \propto (Z - 1), \qquad E_{K\alpha} \approx 10.2\ \text{eV} \cdot (Z-1)^2$$

It was the final proof that the elements are ordered by nuclear charge, not atomic weight – and it showed where elements were missing in the periodic table. Moseley was killed at Gallipoli in 1915, aged 27 (see atomic physics).

In the electron microscope the X-ray spectrum is measured in two ways. An *energy-dispersive* detector (EDS) – a cooled semiconductor crystal – measures the energy of each individual photon and records the whole spectrum at once, with an energy resolution of about 130 eV. A *wavelength-dispersive* spectrometer (WDS) uses Bragg reflection in a crystal and has ten times better resolution, but measures one line at a time; it is the principle of Raimond Castaing's electron microprobe of 1951. Quantitative analysis requires corrections for atomic-number effect, absorption and fluorescence in the specimen (the ZAF correction), and with standards the composition can be determined to within a few per cent from a volume of a cubic micrometre.

Since 1989

- **Aberration correction:** In 1998 Max Haider, Harald Rose and Knut Urban built a corrector of hexapole lenses that gets round Scherzer's theorem – which requires rotational symmetry. The resolution is now below 0.05 nm, and single atoms can be seen, including light ones such as oxygen and lithium.
- **Cryo-electron microscopy:** Biological molecules are frozen in vitreous ice and imaged from thousands of angles. Dubochet, Frank and Henderson received the Nobel Prize in Chemistry in 2017; the method determined the structure of the coronavirus spike protein within weeks in 2020.
- **Silicon drift detectors** (SDD) have replaced the liquid-nitrogen-cooled Si(Li) detectors; they count tens of thousands of times faster and make chemical maps of the specimen routine.
- **Focused ion beam (FIB)** combined with SEM can cut TEM specimens out of a precise location and make 3D reconstructions slice by slice.
- **Desktop models:** SEMs now exist in desktop form for schools and factories.

Further reading

- Joseph I. Goldstein et al.: *Scanning Electron Microscopy and X-Ray Microanalysis* (4th ed., Springer 2018)

- David B. Williams and C. Barry Carter: *Transmission Electron Microscopy* (2nd ed., Springer 2009)
- Ernst Ruska: “The development of the electron microscope and of electron microscopy”, Nobel lecture 1986

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